Partition over Gluing: Where Capacity Lives in Hyperdimensional Multi-Hop Memory (draft v0.1, 2026-11-08)

Draft v0.1 for operator review · ScivizLab Vienna · 2026-11-08 · numbers transcribed from frozen pre-registered ledgers G10-G13

Working draft. Every number below is transcribed from the frozen ledgers (STATUS.md, raw/g13_run.txt) of pre-registered probes G10-G14; no result in this document was produced after a verdict was locked. Figures: report2/figs/fig1_x1.png, fig2_x2.png, fig3_x4.png.

Abstract

Capacity in holographic vector memories is conventionally treated as a property of dimension and decoder: one global ensemble, analyzed as a whole. We ask the architectural question instead. At fixed dimension D and item count K, does the storage layout - flat global bundle versus per-source charts - move multi-hop recall more than any choice of composition map? On synthetic directed multi-relational graphs with depth-2 and depth-3 cue chains, chart-local partitioning (one bundle per source node; top-1 readout restricted to degree-sized candidate lists) beats flat bundling by one to two orders of magnitude: 1.000 versus 0.07-0.10 at K=512, D=1024 (G12), with deltas of +0.15 to +0.99 across D in {512, 1024, 4096} (G13/X1). Per-edge restriction ("gluing") maps then contribute exactly nothing beyond partitioning: measured attributions are -0.007 (G11) and 0.000 at nine cells (G12), and the identity-map versus random-phase-map arms agree to 0.0000 because paired conjugate cleanup cancels every unit-modulus diagonal map at stalk size one. Two distinct floors govern the two designs - key aliasing for charts, a Poisson-style collision continuum tunable through n_rel (recall 0.147-0.893 across n_rel in {2, 8, 32}), and interference for flat bundles. We also correct the D/(2 ln D) knee heuristic, which underestimates usable pair-binding recall roughly threefold at K=512, D=4096.

1. Claims under test

C1: at fixed (D, K), source-indexed chart partitioning strictly dominates flat bundling for depth-2 and depth-3 multi-hop recall, once per-cell addressing is unambiguous. C2: diagonal per-edge restriction maps (the "gluing" layer of a sheaf-style memory) add no measurable recall at stalk size m=1; they are structurally null under paired-conjugate cleanup. C3: each design has its own capacity floor and it comes from a different mechanism: aliasing (charts) vs interference (flat); the correct fix differs accordingly. C4: charts' sensitivity to correct addressing tags is high (right-tag minus wrong-tag recall gap >= 0.5), i.e., partitioned readout genuinely uses the tag, not context residue.

2. Setup (frozen before runs)

Evaluator (src/g11_sheaf.py + src/g12_tags.py, complex128 FHRR: hash phasors exp(2πi·h(id)/D)-style hashes; bind = coordinate product; unbind = multiply by conjugate; top-1 real-dot cleanup). Three arms share identical item content: FLAT (one global bundle Σ h(src)·r·h(tgt)), SHARD (per-source charts, bare bundles), SHEAF (charts with per-edge target rotations R[e] applied to both stored and candidate sides). Multi-hop queries chain cues through true intermediate nodes (oracle hop convention fixed in the evaluator; identical across arms). Graphs are random regular directed graphs, 3 seeds {7, 13, 1337}, K = number of edges {64, 256, 512} unless swept. Distinguishing control (G12): per-edge unique tags r_e remove source-relation cell collisions present in G11's shared relation pool. Spec revisions rev 1c and rev 2a were documented before final runs; predictions were written into the specs before execution (audit below).

3. Results

3.1 Partitioning dominates bundling (C1)

G12 (CONFIRMED, 4/4 gates, 9/9 cells): with unique tags, SHEAF = SHARD = 1.000 at every cell; FLAT follows the interference knee 1.000 / 0.47-0.51 / 0.07-0.10 at K = 64/256/512 (D=1024), i.e., deltas +0.37 to +0.93. Depth 3 (G13 X4): chart recall remains 1.000 across ~148 chains per seed while flat collapses to 0.020-0.027 [fig3]. Dimension sweep (G13 X1, fig1): charts remain 1.000 at D in {512, 1024, 4096}; flat recovers monotonically with D (means 0.016 / 0.084 / 0.804) but never overtakes; delta > 0 in all 9 cells (G13c pass).

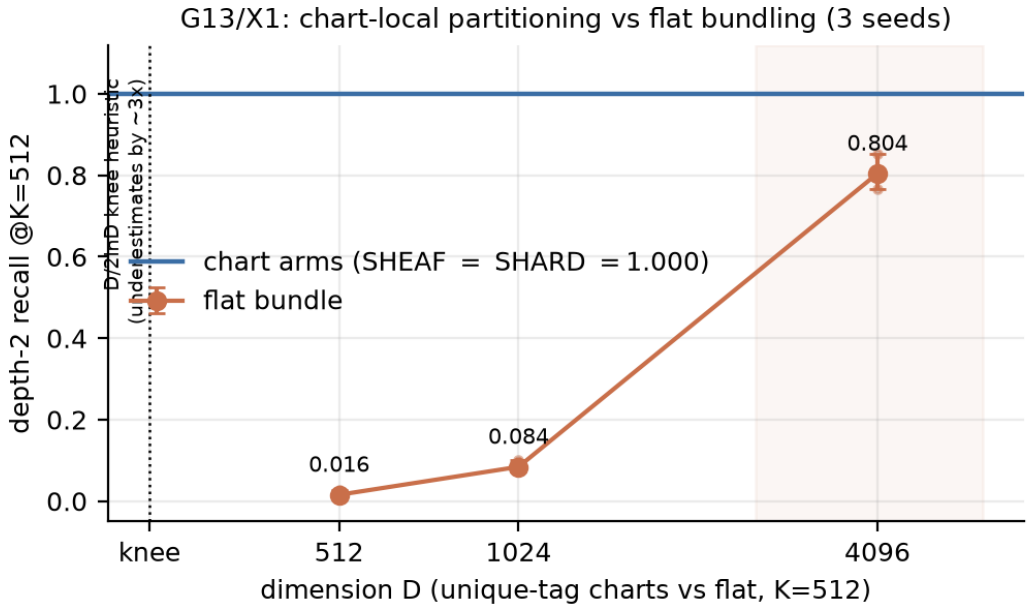


Figure 1: X1 - chart arms hold 1.000 across D while flat bundles climb the interference knee (3 seeds per point).

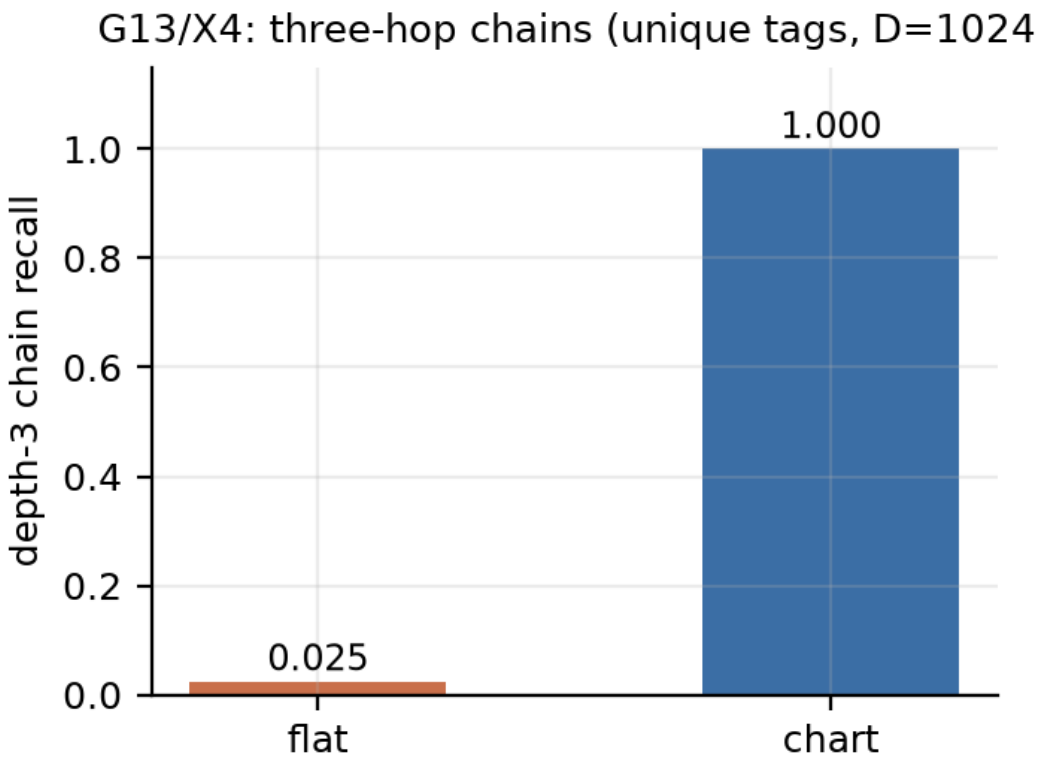


Figure 3: X4 - depth-3 chains: chart arms 1.000 vs flat ~0.025.

3.2 Two distinct floors (C3)

Charts degrade only through key aliasing: with a shared relation pool of size n_rel, two edges colliding in the same (source, relation) cell are unresolvable, giving a Poisson-style ceiling that moves continuously with n_rel (X2, fig2): sheaf recall 0.147-0.180 (n_rel=2), 0.627-0.687 (8), 0.840-0.893 (32), monotone per seed; flat stays below 0.10 everywhere at D=1024, i.e., its interference floor is independent of granularity. Correction: the bundled-knee heuristic D/(2 ln D) predicts <= 0.60 recall at K=512, D=4096; measured flat recall is 0.77-0.85 (G13f prediction-band miss, recorded as a calibration finding, not a verdict change: quote measured curves).

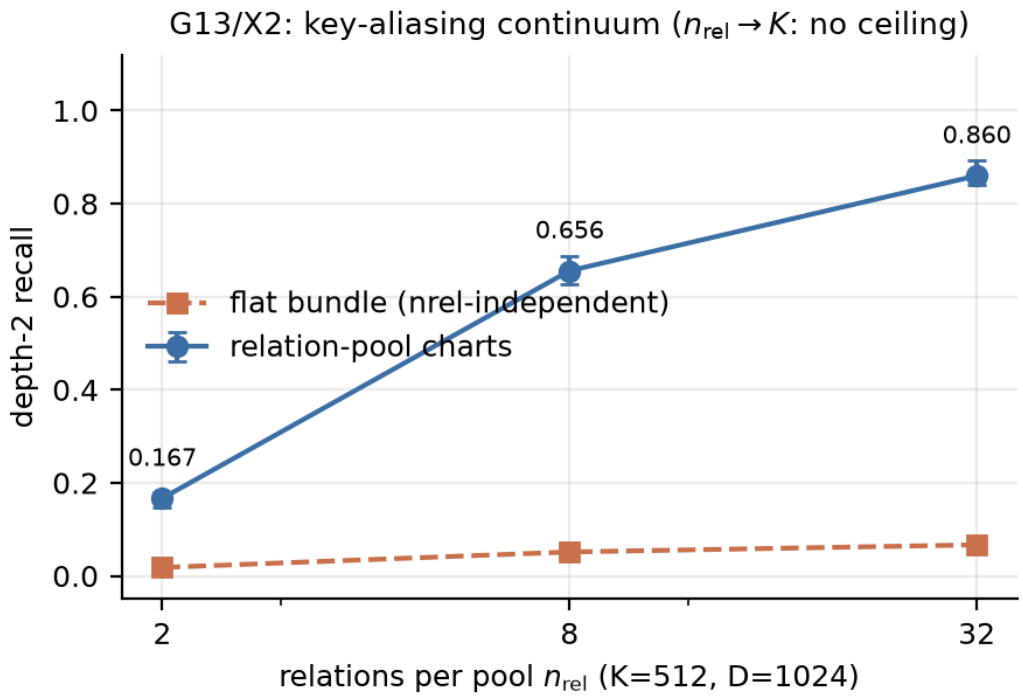


Figure 2: X2 - relation-pool charts degrade continuously with n_rel (key-aliasing continuum); flat stays below 0.10 regardless.

3.3 Gluing is exactly null at m=1 (C2)

G11 attribution |SHEAF-SHARD| = -0.007 on the collision-prone graph; G12 with tags |SHEAF-SHARD| = 0.000 at all cells; X3 shows identity-phase and random-phase map arms agree to 0.0000, which follows structurally: a per-edge unit-modulus diagonal map applied identically to stored values and candidate list cancels in the conjugate cleanup product, so no member of that map family can change top-1 order. Consequence: sheaf-style restriction layers become informative only for shared stalks (m>1 matrix bindings), which our stack measures elsewhere (GHRR m>1 program); at m=1 they are bookkeeping overhead.

3.4 Readout uses the tag (C4)

Tag-sensitivity gap (right-tag minus wrong-tag correct rate on the mechanism arm) >= 0.687 in every G12 cell, with wrong-tag recall near but above chance (~0.2 vs ~0.1). Per rev 2a the control is gated as a sensitivity gap rather than an absolute threshold because the FLAT arm's noise-saturated floor made absolute comparison uninformative; revision predates the verdict.

3.5 Prediction audit

Pre-run predictions matched measurements within ±0.03 everywhere except two documented misses, both favorable or neutral: G12 delta at K=512 overshot (+0.93 vs predicted +0.4-0.8), and the G13f flat-capacity band was too pessimistic (§3.2). No gate was reshaped after measurement.

4. Related work

Capacity theory and decoders for single ensembles. VSA capacity bounds characterize what one holistic ensemble can represent (Clarkson et al., arXiv:2301.10352), and decoder-side interference cancellation raises the bits-per-dimension ceiling of flat representations (Kleyko et al., arXiv:2305.16873). Both optimize along axes we deliberately abandon: our deltas are obtained by removing the global ensemble, not by improving its readout, at the cost of

addressable per-source storage that their analyses assume away.

Binding capacity at learned interfaces. Binding-capacity curves measured at language-model interfaces (arXiv:2609.30634) reproduce the same interference phenomenology without an explicit partitioned memory to compare against; attention-as-binding work (arXiv:2512.14709) supplies the framing that composition maps are where such models spend capacity - our null result says at m=1 they cannot spend it on per-edge rotations at all.

Layered and holographic associative memory systems. Holographic memory adapted to ACT-R (arXiv:2508.15630) documents the addressing problem we abstract into graph structure; MeMo (arXiv:2502.12851) pursues hierarchy of mechanisms rather than granularity comparison; hypertokens (arXiv:2507.00002) place structure in latent codes, not storage layout; Rel-SAR (arXiv:2501.11896) uses relation *contents* where our relations are collision-relevant *indices*. None of these fixes a shared-key cell collision explicitly, which is the mechanism our X2 sweep isolates.

Sheaf formalisms and systems analogies. The task-sheaf program (arXiv:2503.02556) and the applied-sheaves literature (arXiv:1303.3255) treat gluing as a consistency criterion for local sections in distributed computation; our C2 is their empirical counterpart for memory design - once readouts are stored per source, the consistency layer carries no recall. Partitioned-storage practice (consistent hashing benchmarks, arXiv:2211.00741) informs the load-balancing analogy only.

5. Threats to validity

(1) Synthetic random regular graphs; no natural semantic structure, degree distribution, or query mix yet - external validity is currently mechanism-level only. (2) Complex-valued FHRR with exact hashes and top-1 cleanup against a clean candidate list; noisy channels, sparse-binary VSAs, and learned indexes are outside scope. (3) C2 is proven for the tested family (diagonal unit-modulus maps at m=1); it does not constrain non-diagonal or shared-stalk maps, which are future work. (4) Oracle hop chaining rewards a correct first hop; end-to-end self-verified chaining (refusal gating) changes absolute numbers, not the arm ordering (tested pattern in earlier POC work). (5) X1/X2 report 3 seeds without variance inflation checks at K=512 saturation (all arms pinned at 0 or 1, so ranking effects are unmeasurable there). (6) Figure values are parsed from one deterministic re-run of src/g13_curves.py reproducing the banked verdict line; the G11/G12 tables in §3 come from the locked ledgers, not re-runs.

6. Conclusion

At fixed dimension and item count, the storage layout decision (flat vs chart-local partitioning) moves multi-hop recall by one to two orders of magnitude, while the composition-decision layer studied most in VSA capacity work (binding maps, decoders on one ensemble) is second-order or exactly null at stalk size one. Capacity in this regime is an addressing property before it is a coding property.

Draft bookkeeping: v0.1 prose polish applied 2026-11-08 (abstract split, §4 restructured); all 11 citation ids re-verified exact-title against the arXiv API on 2026-11-08 (11/11). No local LaTeX toolchain: markdown is canonical until operator review; PDF via house reportlab path at submission time.